

Hongyi Lin

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EDUCATION

- Ph.D. Candidate in Mechanical Engineering** 9/2022 - 6/2027 (expected)
- **Tsinghua University** Advisor: Prof. Xiaobo Qu
- Dissertation: *Scenario Generation and Perceptual Reasoning Methods for Open-World Autonomous Driving*.
- Two-time recipient of the National Scholarship for Graduate Students (top 2%).
- Visiting Ph.D. Student in Urban Studies and Transportation** 9/2025 - 5/2026
- **Massachusetts Institute of Technology** Advisors: Prof. Jinhua Zhao & Haris Koutsopoulos
- Bachelor of Engineering in Automotive Engineering** 9/2018 - 6/2022
- **South China University of Technology**
- Dissertation: *Analysis of Driving Behavior Characteristics Based on Field Dependence Theory*.
- Highest distinction for undergraduate students, Top 10 Outstanding Student Award (10/20000+).
- Two-time recipient of the National Scholarship for Undergraduate Students (top 1.5%).

RESEARCH PROFILE

Research Interests

- My research focuses on scenario generation and perception for autonomous driving, world models, and the application of VLMs and VLAs to transportation and autonomous-driving systems.
- Additional interests include AI-enabled transportation systems modeling, demand prediction, traffic state prediction and generation, as well as transportation big-data analytics.

Methodological Expertise

- Deep learning, generative modeling, multi-source data fusion, and physics-informed AI modeling.

SELECTED PUBLICATIONS

- [1] **H. Lin**, Y. Liu, S. Li, et al. How Generative Adversarial Networks Promote the Development of Intelligent Transportation Systems: A Survey. *IEEE/CAA Journal of Automatica Sinica*, vol. 10, no. 9, pp. 1781–1796, 2023. (IF=19.2; Rank: 1/89 in Automation & Control Systems; Citations = 118)
- [2] **H. Lin**, W. Shi, H. Huang, et al. Risk-Controllable Multi-View Diffusion for Driving Scenario Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. (**Best Paper Award**)
- [3] **H. Lin**, S. Ming, Y. Liu, et al. AEFusion: An Attention-Based Ensemble Learning Approach for BEV Fusion Perception in Autonomous Modular Buses. *IEEE Transactions on Intelligent Vehicles*, 2024.
- [4] **H. Lin**, Y. Liu, L. Wang, et al. Big Data-Driven Advancements and Future Directions in Vehicle Perception Technologies: From Autonomous Driving to Modular Buses. *IEEE Transactions on Big Data*, 2025.
- [5] **H. Lin**, C. Lyu, Y. He, et al. Enhancing State Representation in Multi-Agent Reinforcement Learning for Platoon-Following Models. *IEEE Transactions on Vehicular Technology*, vol. 73, no. 8, pp. 12110–12114, 2024.
- [6] **H. Lin**, Y. He, Y. Liu, et al. Deep Demand Prediction: An Enhanced Conformer Model with Cold-Start Adaptation for Origin-Destination Ride-Hailing Demand Prediction. *IEEE Intelligent Transportation Systems Magazine*, vol. 16, no. 3, pp. 111–124, 2024.
- [7] **H. Lin**, Y. Liu, L. Wang, et al. High-Precision Calibration and Assessment of Binocular Cameras and LiDAR in Intelligent Vehicles. *IEEE Transactions on Vehicular Technology*, 2025.
- [8] **H. Lin**, Y. He, S. Li, et al. Insights into Travel Pattern Analysis and Demand Prediction: A Data-Driven Approach in Bike-Sharing Systems. *Journal of Transportation Engineering, Part A: Systems*, vol. 150, no. 2, pp. 04023132, 2024.
- [9] S. Zhang*, **H. Lin***, M. Wang, et al. A Dynamic Prompting and Scenario Generation Method for Autonomous Driving Perception via Large-Model Optimization. *Transportation Research Part C: Emerging Technologies*, vol. 188, Article 105672, 2026. *Equal contribution.
- [10] X. Qu, **H. Lin**, Y. Liu. Envisioning the Future of Transportation: Inspiration of ChatGPT and Large Models. *Communications in Transportation Research*, vol. 3, December 2023. (Citations = 140)

- [11] P. Liu, **H. Lin**, Y. Zhao, et al. DesEAD: Enhancing End-to-End Autonomous Driving with Scene Descriptions. In *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, 2025.
- [12] Y. Yang, W. Zhang, **H. Lin**, et al. Applying Masked Language Model for Transport Mode Choice Behavior Prediction. *Transportation Research Part A: Policy and Practice*, vol. 184, June 2024.

Papers Under Review

- [1] **H. Lin**, H. Chen, R. Wang, et al. A NeRF-based Fusion Method for Integrating 3D Road Surface Geometry in Camera-only Object Detection. *IEEE Transactions on Intelligent Vehicles*. Minor Revision.
- [2] **H. Lin**, H. Chen, J. Wang, et al. AiONet: A Unified Vehicle-Infrastructure Cooperative Perception Framework for Urban Mobility Systems. Submitted to *Transportation Research Part C: Emerging Technologies*.
- [3] **H. Lin**, Y. Liu, S. Zhang, et al. VLMs as Selectors Rather than Regressors in Autonomous Systems. Submitted to *Nature Mechanical Engineering*.

SELECTED PATENTS

- [1] X. Qu, **H. Lin**, Y. Liu, et al. A LiDAR-based Perception Method and Devices for Multi-Vehicle Platooning in Intelligent Connected Buses. Chinese Patent, ZL202410433936.7.
- [2] X. Qu, **H. Lin**, Y. Liu, et al. A Camera-based Perception Method and Devices for Multi-Vehicle Platooning in Intelligent Connected Buses. Chinese Patent, ZL202410433937.1.
- [3] X. Qu, **H. Lin**, Y. Liu, et al. A Control Method and Devices for Platooning Docking of Intelligent Connected Buses. Chinese Patent, ZL202410410294.9.
- [4] K. Wang, X. Qu, **H. Lin**. A Vehicle Speed Control Method, Apparatus, Device, and Storage Medium. Chinese Patent, ZL202310639458.0.
- [5] W. Zhang, M. Xu, **H. Lin**, et al. A Connection Device for Modular Cars. Chinese Patent, ZL202420273708.3.

PROJECTS

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| <p>Modular and Autonomous Buses in Sustainable Urban Transportation System</p> <p>Primary student contributor</p> <ul style="list-style-type: none"> - Designed a segmented docking architecture and perception framework for autonomous modular buses in-transit docking. - Proposed a BEV fusion perception approach based on attention mechanisms and ensemble learning, and incorporated road geometric priors from 3D reconstruction into perception models. - Developed demand-prediction methods for large-scale urban transportation demand and cold-start adaptation. - Result: 2 conference/journal papers published and 3 patents authorized. | <p>1/2023 - Present</p> <p><i>Sponsor: National Natural Science Foundation of China</i></p> |
| <p>Autonomous Driving Scenario Generation and 3D Object Detection</p> <p>Primary contributor</p> <ul style="list-style-type: none"> - Proposed a world model for autonomous-driving scenario generation that integrates urban 3D situational data and vehicle-trajectory information. - Developed a unified framework for multi-modal cooperative perception across multiple devices. - Result: 2 conference/journal papers submitted and 1 patent authorized. | <p>3/2024 - Present</p> <p><i>Partner: SAIC Motor Corporation</i></p> |
| <p>VLA-Enabled RoboBus Deployment for Smart Public Mobility</p> <p>Primary student contributor</p> <ul style="list-style-type: none"> - Designed VLA architectures for RoboBus scenario understanding, decision support, and autonomous-driving interaction tasks. - Developed VLA training pipelines using multimodal driving data, scene descriptions, and instruction-action supervision. - Applied model distillation to transfer large-model capabilities into deployable onboard modules for real-time RoboBus applications. | <p>6/2025 - Present</p> <p><i>Sponsor: Hangzhou Airport Economic Zone</i></p> |
| <p>Control of Advanced Driver Assistance Systems for Intelligent Vehicles</p> <p>Project leader</p> <ul style="list-style-type: none"> - Designed and built an autonomous-vehicle experimental platform, implementing localization, perception, decision-making, and control through simulations and real-vehicle experiments. - Result: 2 conference/journal papers published. | <p>5/2020 - 5/2021</p> <p><i>Sponsor: Ministry of Education of China</i></p> |

AWARDS AND HONORS

- CVPR 2026 Best Paper Award, Video Generative Models: Benchmarks and Evaluation 2026
- Tsinghua University Artificial Intelligence Competition, Autonomous Driving Track, First Prize 2025
- National Scholarship for Graduate Students, doctoral level, Ministry of Education of China, top 2% 2025
- National Scholarship for Graduate Students, master's level, Ministry of Education of China, top 2% 2024
- China Robotics and AI Competition, Robot Application Track, First Prize 2024
- China Robotics and AI Competition, Real Car Invitational Competition, Champion 2024
- Outstanding Oral Presentation, Doctoral Forum, Tsinghua University 2024
- Comprehensive First-Class Scholarship, Tsinghua University 2023
- COMAP Interdisciplinary Contest in Modeling (ICM), Finalist 2021
- Top 10 Outstanding Student Award, South China University of Technology (10/20000+) 2020
- National Scholarship for Undergraduate Students, Ministry of Education of China, top 1.5% 2019/2020

PROFESSIONAL ACTIVITIES

Talks and Presentations

- "Risk-Controllable Multi-View Diffusion for Driving Scenario Generation," CVPR, Denver, 2026.
- "VLMs as Selectors Rather than Regressors in Autonomous Systems," MIT SRB Forum, 2026.
- "Bidirectional Physical Field Generation for Low-Altitude-Aware Urban Planning," MIT JTL Lab, 2026.
- "Scenario Generation and Perception-Reasoning for Autonomous Driving," KAIST, 2026.
- "Enhancing Transportation Modeling with Physics-Informed Machine Learning," APTE, National University of Singapore, 2024.
- "Framework for Large-Scale Urban Traffic State Estimation Based on AIGC," KES-STS, Portugal, 2024.
- "Enhancing State Representation in Multi-Agent Reinforcement Learning for Platoon-Following Models," WTC, Qingdao, 2024.

Academic Service

- Session Chair, Smart Transportation and Vehicle Equipment Symposium, Hangzhou, 2024.
- Contributed to editorial and outreach activities for *COMMTR* and *JICV*, including manuscript solicitation, peer review, topical special-issue planning, and international journal promotion.

Reviewer for International Conferences and Journals

- **Journals:** TRA, TRC, TRE, IEEE TIV, IEEE ITSM, IET ITS, COMMTR, JICV.
- **Conferences:** ICML, IEEE IV, IEEE ITSC, WCTR, ITSWC, APTE.

INTERNSHIP EXPERIENCES

Suzhou Automotive Research Institute, Tsinghua University 6/2024 - 8/2024

Research Intern, Autonomous Driving Perception

- Improved the threshold-constraint module of RenderOCC and developed a visualization module compatible with occupancy prediction methods.
- Developed a label-efficient OCC workflow, reducing data-labeling costs by approximately RMB 500,000.
- Result: one journal manuscript submitted and one patent application filed.

TEACHING

Future Transport Modes (Prof. Xiaobo Qu & Kai Wang) Fall 2023 & Fall 2024

Teaching Assistant

- Assisted with course design, course organization, student project mentoring, and assignment evaluation.

REFERENCES

- **Xiaobo Qu**, xiaobo@tsinghua.edu.cn
Elected Member of Academia Europaea; Changjiang Chair Professor with the School of Vehicle and Mobility, THU.
- **Jinhua Zhao**, jinhua@mit.edu
Department Head, DUSP; Professor of Cities and Transportation; Founder, MIT Mobility Initiative; Lead PI, Mens, Manus and Machina (M3S), MIT.